

## GAN ISLAND, Maldives

WMO: 435990

Lat: 0.6905S

Lon: 73.1501E

Elev: 6

StdP: 14.69

Time Zone: 5.00 (E05)

Period: 90-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |       |      |      |       |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-------|------|------|-------|------|-----------------------|------|------|------|--------------------------|------|-------|
|               |            |      | 99.6%                         |       |      | 99%  |       |      | 0.4%                  |      | 1%   |      |                          |      |       |
|               | 99.6%      | 99%  | DP                            | HR    | MCDB | DP   | HR    | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)   | (f)  | (g)  | (h)   | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                      | (o)  | (p)   |
| 12            | 75.2       | 76.0 | 71.8                          | 117.4 | 81.4 | 72.4 | 120.0 | 81.5 | 23.2                  | 79.0 | 20.9 | 80.2 | 8.0                      | 270  | 0.430 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 4             | 7.3                    | 89.0            | 80.4 | 88.6 | 80.2 | 88.1 | 80.0 | 81.8                | 87.0 | 81.5 | 86.7 | 81.0 | 86.4 | 8.9                  | 270  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 80.3                            | 157.1 | 85.7 | 79.8 | 154.9 | 85.4 | 79.3 | 152.0 | 85.0 | 45.6          | 87.5 | 45.1 | 87.0 | 44.7 | 86.9 | 84.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |      |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |      |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |
| 21.5              | 18.6 | 16.3 | N/A                        | N/A  | N/A                | N/A | N/A                                                | N/A  | N/A        | N/A  | N/A        | N/A  | N/A        | N/A  |      |
|                   |      |      | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |      |
|                   |      |      | WB                         | 69.0 | 83.0               | 4.2 | 0.8                                                | 66.0 | 83.6       | 63.5 | 84.1       | 61.1 | 84.6       | 58.0 | 85.1 |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 82.8  | 82.6  | 83.4  | 83.8  | 84.1  | 83.3  | 83.5  | 82.6  | 82.2  | 82.0  | 82.1  | 82.0  | (6)  |      |
| (7)  |                                                                                  | DBStd       | 1.83  | 1.67  | 1.52  | 1.48  | 1.62  | 1.86  | 1.54  | 1.77  | 1.72  | 1.97  | 1.58  | 1.69  | 1.66 | (7)  |
| (8)  |                                                                                  | HDD50       | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | (8)  |
| (9)  |                                                                                  | HDD65       | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | (9)  |
| (10) |                                                                                  | CDD50       | 11970 | 1011  | 934   | 1047  | 1022  | 1032  | 1004  | 1012  | 997   | 961   | 995   | 961   | 994  | (10) |
| (11) |                                                                                  | CDD65       | 6495  | 546   | 514   | 582   | 572   | 567   | 554   | 547   | 532   | 511   | 530   | 511   | 529  | (11) |
| (12) |                                                                                  | CDH74       | 76950 | 6349  | 6281  | 7292  | 7245  | 7017  | 6774  | 6425  | 6112  | 5781  | 5983  | 5763  | 5928 | (12) |
| (13) |                                                                                  | CDH80       | 26609 | 2087  | 2341  | 2922  | 2998  | 2672  | 2563  | 2175  | 1922  | 1749  | 1766  | 1699  | 1715 | (13) |
| (14) | Wind                                                                             | WSAvg       | 7.5   | 6.1   | 6.2   | 5.2   | 8.1   | 9.3   | 7.3   | 7.0   | 7.1   | 8.1   | 8.9   | 9.2   | 7.6  | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 93.7  | 8.7   | 4.4   | 5.6   | 6.6   | 9.3   | 7.1   | 7.0   | 7.7   | 8.6   | 10.7  | 7.7   | 10.1 | (15) |
| (16) |                                                                                  | PrecMax     | 125.2 | 21.2  | 11.1  | 14.5  | 12.8  | 20.6  | 14.0  | 14.0  | 14.8  | 15.1  | 28.2  | 18.8  | 22.1 | (16) |
| (17) |                                                                                  | PrecMin     | 61.0  | 0.3   | 0.3   | 0.4   | 2.1   | 2.8   | 0.2   | 0.5   | 2.1   | 2.2   | 1.3   | 0.6   | 3.4  | (17) |
| (18) |                                                                                  | PrecStd     | 15.6  | 5.0   | 3.4   | 3.4   | 2.9   | 4.8   | 4.0   | 3.2   | 3.6   | 3.5   | 5.5   | 4.8   | 5.4  | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 87.9  | 88.7  | 89.6  | 89.8  | 89.0  | 89.1  | 89.0  | 88.4  | 88.2  | 88.0  | 88.0  | 87.8 | (19) |
| (20) |                                                                                  |             | MCWB  | 79.7  | 80.1  | 80.4  | 80.8  | 80.6  | 80.5  | 80.1  | 79.9  | 79.9  | 79.6  | 80.0  | 79.9 | (20) |
| (21) |                                                                                  | 2%          | DB    | 87.1  | 87.9  | 88.8  | 89.0  | 88.2  | 88.3  | 88.0  | 87.5  | 87.2  | 87.1  | 87.1  | 86.9 | (21) |
| (22) |                                                                                  |             | MCWB  | 79.3  | 79.8  | 80.1  | 80.5  | 80.4  | 80.1  | 79.8  | 79.6  | 79.6  | 79.3  | 79.7  | 79.5 | (22) |
| (23) |                                                                                  | 5%          | DB    | 86.5  | 87.3  | 88.1  | 88.4  | 87.6  | 87.8  | 87.2  | 86.9  | 86.5  | 86.4  | 86.4  | 86.2 | (23) |
| (24) |                                                                                  |             | MCWB  | 79.0  | 79.5  | 79.8  | 80.3  | 80.3  | 79.9  | 79.6  | 79.3  | 79.3  | 79.0  | 79.5  | 79.3 | (24) |
| (25) |                                                                                  | 10%         | DB    | 86.0  | 86.7  | 87.5  | 87.8  | 86.9  | 87.1  | 86.5  | 86.1  | 85.8  | 85.8  | 85.7  | 85.5 | (25) |
| (26) |                                                                                  |             | MCWB  | 78.7  | 79.2  | 79.6  | 80.1  | 80.1  | 79.7  | 79.3  | 79.0  | 79.1  | 78.8  | 79.2  | 79.0 | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 81.1  | 81.8  | 82.2  | 82.6  | 82.2  | 81.9  | 81.7  | 81.5  | 81.7  | 80.9  | 81.6  | 81.2 | (27) |
| (28) |                                                                                  |             | MADB  | 86.1  | 86.4  | 87.7  | 87.9  | 87.2  | 87.2  | 86.8  | 86.4  | 86.1  | 86.0  | 85.7  | 85.9 | (28) |
| (29) |                                                                                  | 2%          | WB    | 80.4  | 81.0  | 81.4  | 81.7  | 81.5  | 81.2  | 80.8  | 80.6  | 80.7  | 80.3  | 80.7  | 80.5 | (29) |
| (30) |                                                                                  |             | MADB  | 85.6  | 86.2  | 87.0  | 87.4  | 86.7  | 86.7  | 86.3  | 86.1  | 85.8  | 85.3  | 85.7  | 85.5 | (30) |
| (31) |                                                                                  | 5%          | WB    | 79.9  | 80.4  | 80.8  | 81.0  | 80.9  | 80.7  | 80.2  | 80.0  | 80.0  | 79.8  | 80.1  | 80.0 | (31) |
| (32) |                                                                                  |             | MADB  | 85.1  | 85.9  | 86.6  | 86.9  | 86.3  | 86.3  | 85.8  | 85.4  | 85.1  | 84.9  | 85.1  | 84.8 | (32) |
| (33) |                                                                                  | 10%         | WB    | 79.3  | 79.9  | 80.2  | 80.6  | 80.5  | 80.1  | 79.8  | 79.5  | 79.5  | 79.2  | 79.6  | 79.5 | (33) |
| (34) |                                                                                  |             | MADB  | 84.5  | 85.4  | 86.0  | 86.6  | 85.9  | 85.6  | 85.2  | 84.8  | 84.5  | 84.2  | 84.5  | 84.2 | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR  | 6.4   | 6.4   | 7.4   | 7.3   | 7.1   | 7.0   | 7.3   | 7.3   | 7.0   | 7.1   | 6.7   | 6.6  | (35) |
| (36) |                                                                                  |             | MCDBR | 6.6   | 6.8   | 7.9   | 7.7   | 7.5   | 7.5   | 7.8   | 7.9   | 7.7   | 7.5   | 7.2   | 7.0  | (36) |
| (37) |                                                                                  |             | MCWBR | 3.1   | 3.2   | 3.4   | 3.5   | 3.5   | 3.3   | 3.5   | 3.6   | 3.6   | 3.7   | 3.8   | 3.4  | (37) |
| (38) |                                                                                  | 5% WB       | MCDBR | 6.7   | 6.7   | 7.6   | 7.6   | 7.1   | 7.1   | 7.5   | 7.4   | 7.4   | 7.5   | 6.9   | 6.7  | (38) |
| (39) |                                                                                  |             | MCWBR | 3.4   | 3.5   | 3.7   | 3.8   | 3.7   | 3.5   | 3.8   | 3.7   | 3.9   | 4.0   | 3.9   | 3.6  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.407 | 0.412 | 0.409 | 0.399 | 0.400 | 0.415 | 0.427 | 0.413 | 0.405 | 0.405 | 0.397 | 0.395 | (40) |      |
| (41) |                                                                                  | taud        | 2.530 | 2.494 | 2.519 | 2.579 | 2.590 | 2.527 | 2.480 | 2.530 | 2.545 | 2.554 | 2.579 | 2.581 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 293   | 292   | 290   | 286   | 279   | 270   | 269   | 279   | 287   | 291   | 294   | 296   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 35    | 36    | 35    | 32    | 31    | 32    | 34    | 33    | 34    | 34    | 33    | 33    | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 1881  | 2036  | 2098  | 1929  | 1736  | 1669  | 1578  | 1696  | 1766  | 1827  | 1770  | 1784  | (44) |      |
| (45) |                                                                                  | RadStd      | 142   | 116   | 125   | 124   | 89    | 131   | 78    | 112   | 134   | 143   | 143   | 143   | 149  | (45) |

## Historical Trends

|                             | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-----------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                             |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                             |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (0 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page